

QUOR: An AI-Assisted Next-Generation Firewall Integrating Deep Packet Inspection, Flow-Based Intrusion Detection and Honeypot Monitoring

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Abstract—The increasing complexity of modern cyber threats has made traditional rule-based firewalls insufficient for protecting network infrastructures. This paper presents QUOR, an AI-assisted next-generation firewall that integrates deep packet inspection, flow-based intrusion detection, ransomware detection, honeypot monitoring, and encrypted traffic analysis. The system is designed using a modular architecture that enables real-time packet capture, rule enforcement, traffic inspection, and anomaly detection using machine learning models. QUOR also incorporates Explainable Artificial Intelligence (XAI) techniques to improve transparency in intrusion detection decisions and utilizes mitmproxy-based monitoring to detect potential Server Name Indication (SNI) spoofing attacks in encrypted communications. Experimental analysis demonstrates that combining multiple security layers significantly improves the detection of suspicious traffic patterns and advanced cyber threats. The proposed framework provides a flexible and extensible platform for building intelligent firewall systems capable of securing modern network environments.

Index Terms—Next Generation Firewall, Intrusion Detection System, Deep Packet Inspection, Machine Learning, Honeypot, Ransomware Detection, Network Security

I. INTRODUCTION

Modern network infrastructures are exposed to increasingly sophisticated cyber threats including ransomware attacks, encrypted malware communication, and advanced persistent threats. Traditional firewalls rely primarily on static rule-based filtering mechanisms that analyze network packets based on IP addresses, ports, and protocols. Although these mechanisms provide basic protection, they are unable to effectively detect advanced attack techniques such as polymorphic malware, encrypted command-and-control traffic, and application-layer exploits.

Next-Generation Firewalls (NGFWs) extend traditional firewall functionality by integrating deep packet inspection, intrusion detection systems, and application-level traffic analysis. However, many commercial firewall platforms are proprietary, expensive, and difficult to customize for experimental security research.

To address these limitations, this paper introduces QUOR, a modular next-generation firewall framework designed to integrate multiple security technologies into a unified architecture. The system combines rule-based packet filtering with deep packet inspection, machine learning-based flow intrusion

detection, honeypot monitoring, ransomware detection, and encrypted traffic analysis. In addition, QUOR incorporates explainable artificial intelligence techniques to enhance the interpretability of machine learning decisions. The proposed architecture provides an extensible platform for developing intelligent firewall systems capable of detecting emerging cyber threats.

II. RELATED WORK

Several research efforts have explored advanced firewall technologies and intrusion detection mechanisms to address modern cybersecurity challenges.

Recent studies on cloud-based next-generation firewalls have demonstrated the importance of integrating deep packet inspection and application awareness to handle encrypted and dynamic network traffic. Other research has focused on detecting deep packet inspection evasion techniques by analyzing contextual packet sequences rather than individual packets.

Hybrid machine learning approaches have also been proposed to improve intrusion detection accuracy by combining supervised and unsupervised learning models. Reinforcement learning-based firewall optimization techniques have been investigated to dynamically adjust firewall policies based on network traffic behavior.

Another important area of research focuses on monitoring encrypted traffic using selective TLS decryption or passive monitoring techniques. These approaches aim to improve network visibility without compromising privacy or introducing significant performance overhead.

In addition to academic research, commercial firewall systems such as Fortinet FortiGate and Cisco Secure Firewall provide enterprise-grade NGFW solutions that integrate intrusion prevention, deep packet inspection, and threat intelligence services. However, these systems often rely on proprietary architectures and limited customization capabilities, which motivates the development of open and modular firewall frameworks such as QUOR.

III. SYSTEM ARCHITECTURE

The QUOR firewall system follows a modular architecture where network traffic passes through several security layers before being allowed or blocked. Each module performs a

specific function, enabling flexible deployment and easier system extensibility.

The main components of the QUOR architecture include packet capture, rule-based filtering, deep packet inspection, flow-based intrusion detection, ransomware detection, honeypot monitoring, encrypted traffic analysis, and logging.

A. Packet Capture and Traffic Monitoring

The packet capture module monitors network traffic in real time using the Scapy library. It captures incoming and outgoing packets from the network interface and extracts key attributes such as source and destination IP addresses, protocol type, port numbers, packet size, and TCP flags. These features are forwarded to subsequent modules for further inspection.

B. Firewall Rule Enforcement

The firewall rule engine acts as the first security layer in the system. It applies predefined filtering policies to block traffic associated with malicious IP addresses, restricted ports, or abnormal session states. Packets that satisfy the rule conditions are forwarded to deeper inspection modules.

C. Deep Packet Inspection

Deep Packet Inspection (DPI) examines packet payloads in addition to header information. This enables the firewall to detect malicious content embedded in application-layer traffic such as HTTP requests, DNS queries, and file transfer sessions. DPI improves threat detection accuracy by identifying abnormal protocol behavior and suspicious payload patterns.

D. Flow-Based Intrusion Detection System

The Flow IDS groups packets into flows based on the standard five-tuple network parameters: source IP, destination IP, source port, destination port, and protocol. Statistical features such as packet counts, flow duration, and average packet size are extracted from each flow and analyzed using a machine learning model. This enables the system to detect anomalous traffic patterns associated with cyber attacks.

E. Ransomware Detection

Ransomware detection is implemented using behavioral traffic analysis techniques. The system monitors abnormal communication patterns such as rapid file transfer requests, suspicious encryption-related traffic, and unusual outbound connections. Machine learning models are used to classify suspicious behavior and generate alerts when ransomware activity is suspected.

F. Honeypot Monitoring

The honeypot module provides a controlled environment that simulates vulnerable services. Suspicious traffic can be redirected to the honeypot, allowing the system to observe attacker behavior and collect threat intelligence. This information can be used to update detection models and improve firewall policies.

G. Encrypted Traffic Monitoring and SNI Analysis

Since a large portion of internet traffic is encrypted, QUOR monitors TLS handshakes to extract Server Name Indication (SNI) information. The system uses mitmproxy-based traffic analysis to detect anomalies such as domain spoofing or suspicious encrypted communication patterns.

H. Explainable AI for Threat Analysis

To improve transparency in intrusion detection, QUOR incorporates explainable artificial intelligence techniques that help interpret machine learning decisions. These techniques allow security administrators to understand why specific traffic flows were classified as malicious, improving trust in automated security systems.

IV. TECHNICAL IMPLEMENTATION

The QUOR firewall system is implemented primarily using Python and integrates several open-source networking tools. Packet capture and manipulation are performed using the Scapy library, while machine learning models are implemented using standard Python data science frameworks.

Traffic flows are generated from captured packets and converted into feature vectors used by the intrusion detection model. These features are normalized before being processed by the machine learning classifier.

Mitmproxy is used for monitoring encrypted TLS sessions and extracting SNI metadata, while the honeypot environment is configured to simulate vulnerable services for capturing attacker interactions. Security events generated by each module are recorded in a centralized logging system for monitoring and analysis.

V. CONCLUSION

This paper presented QUOR, an AI-assisted next-generation firewall framework designed to enhance network security through the integration of deep packet inspection, flow-based intrusion detection, ransomware detection, honeypot monitoring, and encrypted traffic analysis. The modular architecture enables flexible deployment and real-time monitoring of network traffic while maintaining scalability and extensibility. Future work will focus on improving machine learning models, expanding encrypted traffic analysis capabilities, and deploying the system in large-scale cloud environments.

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